Jianhui YAN

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https://yimkf.github.io/

EDUCATION

Research Assistant 01/2025 – present
MEI Lab at the City University of Hong Kong Hong Kong, China
Research advisor: Prof. Kening Zhu

Research Intern 06/2024 – 09/2024
Human Computer Interaction Lab at Saarland University Saarbrücken, Germany
Research advisor: Prof. Jürgen Steimle

M.S. Electronic Information Engineering 09/2022 – present
School of Electronic and Information Engineering, South China University of Technology Guangzhou, China
Research advisor: Prof. Lin Shu

B.S. Information Engineering 09/2018 – 06/2022
School of Electronic and Information Engineering, South China University of Technology Guangzhou, China
GPA: 3.71/4.0

RESEARCH INTERESTS

Human Computer Interaction (HCI), Virtual/Augmented Reality (VR/AR), Haptic,Wearable System

PUBLICATION

Under Review

EMS Hand Prop: Leveraging the Loss of Sense of Agency Caused by Electrical Muscle Stimulation to Enhance the Immersion of Hand-as-prop Interactions

Jianhui Yan, Jiesi Zhang, Haoqiang Hua, Wenxuan Wu, Hongnan Lin, Qiwei Xiong, Jianxiu Jin, Lin Shu.
Submitted to the **International Journal of Human-Computer Studies**

HapticPipette: Direct Manipulation for Customizing Touch Sensations in Mixed-Reality Environments

Arata Jingu, **Jianhui Yan**, Maja Fehlberg, Roland Bennewitz, Jürgen Steimle.

Accepted

VirCHEW Reality: On-Face Kinesthetic Feedback for Enhancing Food-Intake Experience in Virtual Reality

Qingqin Liu, Ziqi Fang, Jiayi Wu, Shaoyu Cai, **Jianhui Yan**, and Kening Zhu
Contionally Accepted by **SIGGRAPH 2025**

RESEARCH EXPERIENCE

[1] EMS Hand Prop: Leveraging the Loss of Sense of Agency Caused by Electrical Muscle Stimulation to Enhance the Immersion of Hand-as-prop Interactions 05/2023 – 01/2024

- **Novel Concept:** Repurposing the conventional drawback of electrical muscle stimulation (EMS), the loss of SoA, to enhance the immersion of hand-as-prop interactions.
- **Interactive System Integration**: EMS Hand Prop, a system combining EMS actuation and data-glove-based gesture recognition, enables object retrieval through non-dominant hand gestures while interacting with these objects using the dominant hand.
- **Empirical Study:** Designed two within-subject user studies (n = 12) and conducted quantitative and qualitative analysis to evaluate the EMS Hand Prop.

[2] HapticPipette: Direct Manipulation for Customizing Touch Sensations in Mixed-Reality Environments 06/2024 – 09/2024

- **MR Interaction Technique Development:** Developed an MR real-time interaction technique that captures haptic properties, assigns the properties to virtual objects, and perceives them through finger-worn devices.
- **System Integration:** Established the communication between Quest 3, haptic devices, and mobile computers, enabling a fully mobile system.
- **Ideation and Application**: Contributed to brainstorming and ideation for the paper, exploring multiple application possibilities and implementing them.

[3] Smart Surface: Haptic Feedback on the Touch-screen Control Panel in Car 03/2025 – present

- **Ultrasound Haptic Prototype:** Built an ultrasound array to render haptic feedback based on SonicSurface

[4] VirCHEW Reality: On-Face Kinesthetic Feedback for Enhancing Food-Intake Experience in Virtual Reality 01/2025

- **Data Analysis**: Processed and synchronized the surface electromyogram (sEMG) data based on the interaction in virtual reality.
- **Application Implementation**: Implemented the application scenarios in virtual reality.

[5] Construction of Immersive and Interactive Methodology based on Physiological Indicators to Subjectively and Objectively Assess Comfort and Performances in Work Offices 03/2024 – 05/2024

- **Data Analysis**: Processed and synchronized the physiological data, and analyzed the data using neurokit2 .
- **User Study Implementation**: Recreated a real-life room in Unity and set up the environment for the user study

AWARDS & SCHOLARSHIP

National Encouragement Scholarship 2020

The Second Prize Scholarship of South China University of Technology 2019

Second Prize of Guangdong Province of National Undergraduate Mathematical Contest in Modeling 09/2020

Second Prize of Guangdong Undergraduate Electronic Design Competition 11/2020

"Hongping Evergreen Fund" Student Science and Technology Innovation Third-class Scholarship (2 items) 2020

SKILLS & LANGUAGE

Hardware & Software Skills

- Software: Unity3D(C#), AR/VR, Python, Git, Matlab, Machine Learning, Data Analysis
- Hardware: Electrical Muscle Stimulation, Arduino, Wearable Systems, PlatformlO, PCB, Ultrasound haptic

Language:
IELTS: 7

TEACHING EXPERIENCE & SERVICE

EuroHaptics 2024 07/2024
Student Volunteer Lille, France

Digital system design 09/2023 – 12/2023
Teaching Assistant South China University of Technology